

M339D: February 16th, 2026.

European Put Options.

Usually, a right but not an obligation to SELL an underlying @ the strike price!

At time $t=0$: The writer and the buyer of the put agree on:

- the underlying asset: $S(t)$, $t \geq 0$;
- the exercise date T ;
- the strike/exercise price K

The put premium $V_p(0)$ is paid by the put's buyer to the put's writer.

At time T :

- The put's owner has a right but not an obligation to sell one unit.
- The put's writer is obligated to do what the put's owner decides.

The put owner's optimal behavior is:

IF $S(T) < K$, then exercise.

IF $S(T) \geq K$, then do not exercise.

PAYOFF:
 $K - S(T)$

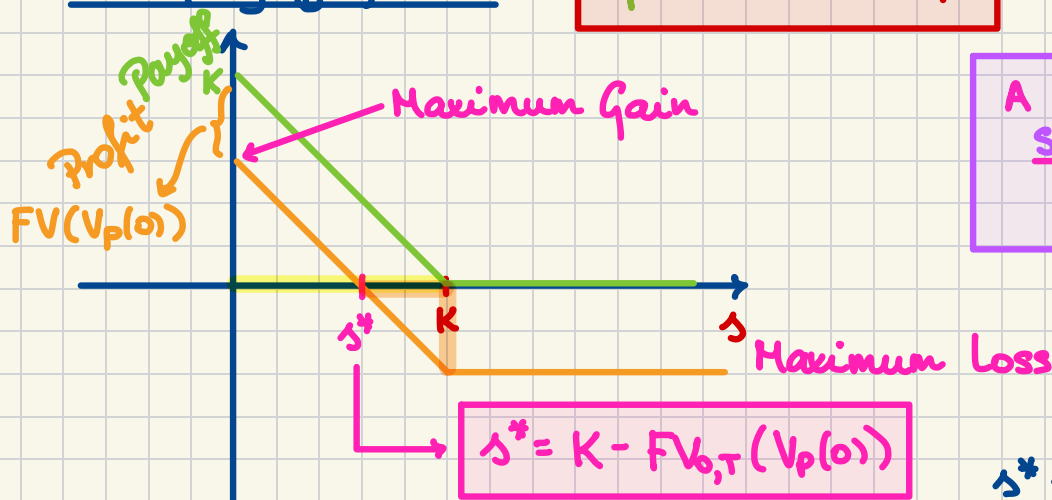
0

The payoff:

$$V_p(T) = \max(K - S(T), 0) = (K - S(T))_+$$

The payoff f'tion:

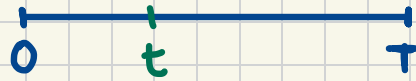
$$v_p(s) = (K - s)_+$$



A LONG PUT IS A SHORT POSITION w.r.t. the underlying.

Moneyiness.

Consider an option written @ time 0 w/ an exercise date T.



Imagine the cashflow that would happen if the option were exercised @ time t.

e.g.,
call $\frac{S(t) - K}{K - S(t)}$
put

If the cashflow is $\begin{cases} > 0, & \text{the option is in-the-money} \\ = 0, & \text{the option is @-the-money} \\ < 0, & \text{the option is out of the money} \end{cases}$

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Problem Set #6

European put options.

Problem 6.1. The initial price of the market index is \$900. After 3 months the market index is priced at \$915. The nominal rate of interest convertible monthly is 4.8%. The premium on the put, with a strike price of \$930, is \$8.00. What is the profit at expiration for a long put?

- ☹ (a) \$15.00 loss
 (b) \$6.90 loss
 (c) \$6.90 gain ✓
 ☹ (d) \$15.00 gain
 (e) None of the above.

DISTRACTION! $i^{(12)}$

$\Rightarrow$ effective monthly
 $j = \frac{i^{(12)}}{12} = 0.004$

→:

$$FV_{0,T}(V_P(0)) = 8 \cdot (1.004)^3$$

$$\text{Payoff} = (K - S(T))_+ = (930 - 915)_+ = 15$$

$$\text{Profit} = 15 - 8(1.004)^3 = \underline{6.90}$$



Problem 6.2. Sample FM(DM) #12

Consider a European put option on a stock index without dividends, with 6 months to expiration, and a strike price of 1,000. Suppose that the nominal annual risk-free rate is 4% convertible semiannually, and that the put costs 74.20 now. What price must the index be in 6 months so that being long the put would produce the same profit as being short the put?

- ☹ i.r. A. 922.83
 ☹ PAYOFF B. 924.32
 ☹ CALL C. 1,000.00
 ☹ CALL+i.r. D. 1,075.68
 E. 1,077.17

We're really looking for the break-even price.

$$\begin{aligned}
 S^* &= K - FV_{0,T}(V_P(0)) \\
 &= 1000 - 74.20(1.02) = \underline{924.32}
 \end{aligned}$$

effective per half-year is

$$j = \frac{0.04}{2} = 0.02$$



Problem 6.3. Farmer Shaun is producing sweet potatoes. He intends to harvest 10,000—cartons' worth in six months. His total costs are \$12.00 per carton.

He wishes to hedge using European put options. There are two puts on sweet potatoes with the exercise date in six months available: one with the strike price of \$13 per carton and another with the strike price of \$15 per carton. Their premiums are \$0.15 and \$0.18, respectively.

Assume that the prevailing risk-free interest rate is 4% effective for the half-year period.

At harvest time, in six months, it turns out that the sweet-potato spot price equals \$14. What would Farmer Shaun's profit be if he had decided to hedge using the \$13-strike put versus his profit if he had decided to use the \$15-strike put to hedge?

focus on the payoff w/out production costs.

unhedged : $S(T)$

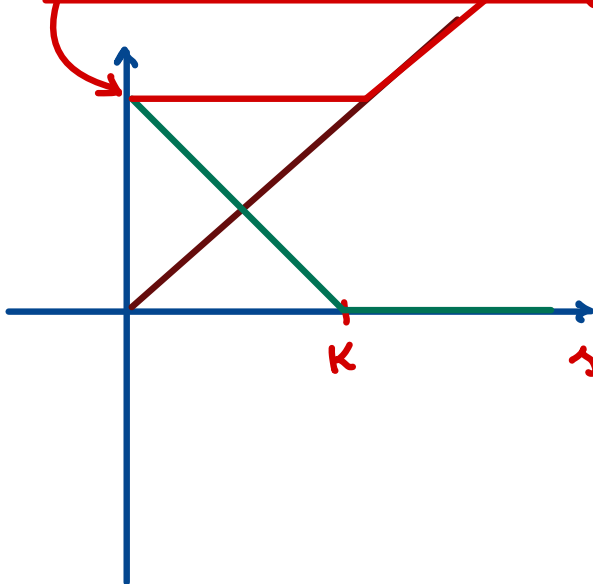
hedge : $(K - S(T))_+$

total hedged :

$$S(T) + (K - S(T))_+ = \begin{cases} K & \text{if } K > S(T) \\ S(T) & \text{if } K \leq S(T) \end{cases}$$

$$= \max(S(T), K)$$

FLOOR = long underlying + long put



The total profit:

$$\begin{aligned} & 10,000 \cdot \max(S(T), K) - 10,000 \cdot 12 - 10,000 \cdot V_p(0, K) \cdot (1.04) = \\ & = 10,000 (\max(S(T), K) - 12 - V_p(0, K) \cdot (1.04)) \end{aligned}$$

For $K = 13$: _____

For $K = 15$: